

NITSCI: Cross-Expert Lesion-Aware Attention with Ordinal Dual Heads and Cross-Dataset Transfer for Diabetic Retinopathy Grading

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Abstract

Background. Grading diabetic retinopathy (DR) from colour fundus photographs on the five-level international scale is difficult for deep networks when the target dataset is small: the cues that separate adjacent grades are a few small lesions, the grades are ordered, and the classes are imbalanced.

Methods. We present NITSCI, a grading network that couples an EfficientNetV2-S backbone with two token-level experts: a global expert (a Transformer encoder layer over all spatial tokens) and a lesion expert (the same tokens re-weighted by a learned, sparsity-regularised lesion-attention map). The experts exchange information through bidirectional cross-attention and are merged by an image-adaptive gate. A dual head predicts class probabilities and a continuous ordinal grade, which are combined into one score. The model is pretrained on APTOS-2019 ([**naptos**] images) and fine-tuned on IDRiD with five-fold cross-validation; decision thresholds are fitted on out-of-fold predictions only, and the five fold models are ensembled with flip test-time augmentation. All results are reported on a stratified held-out IDRiD test set ([**intest**] images) that is never used for training, model selection or threshold fitting.

Results. On five-class grading NITSCI reaches an accuracy of [**acc5**]% (95% CI [**acc5ci**]) and a quadratic weighted kappa of [**qwk5**] (95% CI [**qwk5ci**]); [**within1**]% of test images are graded within one level of the reference. For referable DR (grade ≥ 2) it obtains an AUC of [**refauc**], sensitivity [**refsens**]% and specificity [**refspec**]%. Ablations quantify the contribution of APTOS pretraining, the lesion expert, cross-attention, the gate, the ordinal head and the inference-time ensemble. The model has 21.25 M parameters and needs 23.0 GFLOPs per 448×448 image.

Conclusions. [AUTHOR: One sentence stating the main finding once results are filled, e.g. which components matter most, followed by: "External validation on independent cohorts is required before any clinical use."]

Keywords: diabetic retinopathy; fundus image grading; lesion-aware attention; cross-attention; mixture of experts; ordinal regression; transfer learning; IDRiD

Highlights

- A global expert and a lesion expert exchange information through bidirectional cross-attention.
- An image-adaptive gate weighs global context against lesion evidence for each image.
- Classification and ordinal regression heads are fused into one continuous grade score.
- Thresholds are fitted on out-of-fold data; the test set is never used for any choice.
- Bootstrap confidence intervals and paired McNemar tests accompany every comparison.

1 Introduction

Diabetic retinopathy is a leading cause of preventable vision loss in working-age adults; about one in five people with diabetes is estimated to have some degree of DR, and the absolute number

of affected people is projected to keep rising through 2045 [1]. Screening with colour fundus photography is effective, but grading every image by a trained reader does not scale to the size of the diabetic population. Deep learning systems have reached specialist-level performance for detecting referable DR when trained on very large private datasets [2–4], and an autonomous system has been evaluated in a prospective pivotal trial in primary care [5].

The situation is different when only a small public dataset is available. The Indian Diabetic Retinopathy Image Dataset (IDRiD) [6, 7], for example, provides a few hundred graded images. Three properties of the task then become limiting. *First*, adjacent grades on the international scale [8] are separated by small, sparse findings: a single microaneurysm moves an image from grade 0 to grade 1, and the count and location of haemorrhages decide between grades 2 and 3. A network that pools the whole image uniformly can dilute this evidence. *Second*, the labels are ordered, so predicting grade 4 for a grade-0 image is a much worse error than predicting grade 1, yet standard cross-entropy treats both equally. *Third*, the grade distribution is imbalanced, and small datasets give unstable estimates of both the weights and the decision rule.

Prior work addresses these issues separately: lesion-focused models zoom into suspicious regions [9], attention modules relate disease-specific features [10], ordinal formulations reduce large errors [11, 12], and pretraining on larger fundus collections or foundation models improves data efficiency [13]. In this paper we combine these ideas in one compact model and, equally importantly, in one evaluation protocol that keeps the test set untouched.

Contributions.

1. **Cross-expert lesion-aware architecture.** We propose a head that splits backbone tokens into a global expert and a lesion expert, lets each attend to the other, and fuses them with an image-adaptive gate in the spirit of mixtures of experts [14, 15]. It adds 1.40 M parameters to the backbone.
2. **Ordinal dual-head decision.** A classification head and a regression head are trained jointly and combined into one continuous grade score whose thresholds are fitted on out-of-fold predictions.
3. **Leakage-free protocol for small datasets.** Cross-dataset pretraining on APTOS-2019 [16], five-fold fine-tuning, a fold-ensemble with test-time augmentation, and a held-out test set that is used once, with bootstrap confidence intervals and paired significance tests.
4. **Component-level evidence.** Architecture, data and inference ablations under identical splits, reported for both five-class grading and referable-DR screening.

The remainder of the paper reviews related work (Section 2), describes the data (Section 3) and the method (Section 4), details the experimental setup (Section 5), reports results (Section 6), and discusses findings and limitations (Section 7).

2 Related work

Deep learning for DR grading. Early convolutional networks for DR classification [17], and later residual and Inception-style backbones [18, 19], were followed by large-scale systems trained on tens of thousands of graded images [2, 3] and by systems covering the full disease spectrum [4]. Public benchmarks include Messidor [20], DDR [21], APTOS-2019 [16] and IDRiD [6, 7]. More recently, retinal foundation models pretrained with self-supervision on large unlabelled collections have improved label efficiency [13]. Our work targets the complementary setting of a compact supervised model fine-tuned on a small dataset.

Lesion-aware and attention-based grading. Zoom-in-Net [9] generates attention maps that localise suspicious regions and re-examines them at higher resolution, and CANet [10] uses attention to model relations between DR and diabetic macular oedema features. Transformers [22–24]

Table 1: Grade distribution of the datasets used. The IDRiD test split is a stratified 15% hold-out.

Dataset	Grade 0	Grade 1	Grade 2	Grade 3	Grade 4	Total
APTOS-2019 (pretraining)	[nap0]	[nap1]	[nap2]	[nap3]	[nap4]	[naptos]
IDRiD (all)	[nid0]	[nid1]	[nid2]	[nid3]	[nid4]	[nidrid]
IDRiD test (held out)	[nte0]	[nte1]	[nte2]	[nte3]	[nte4]	[ntest]

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Grade histograms of IDRiD and APTOS-2019 (export cell).

Figure 1: Grade distributions of IDRiD (target) and APTOS-2019 (pretraining).

model long-range dependencies between image regions. Pixel-level lesion segmentation methods [25] provide explicit lesion maps but need lesion annotations. NITSCI learns a lesion-attention map *without* lesion masks, using only grade labels and a sparsity prior, and uses it to define a second token stream rather than only to re-weight a single pooled feature.

Ordinal formulations. Ordinal regression methods exploit label order, e.g. by decomposing the problem into binary rank tasks [11] with rank-consistent outputs [12]. A simpler and widely used alternative in DR grading is to regress the grade and to optimise the cut-points between grades; quadratic weighted kappa [26] is the standard metric that rewards this behaviour. We combine a regression output with the expected grade of the classification head.

Transfer, ensembling and pre-processing. ImageNet pretraining [27], including ImageNet-21k [28], is the usual starting point; an intermediate stage on a larger in-domain dataset narrows the domain gap. Deep ensembles [29] and weight averaging [30] improve accuracy and calibration. Ben Graham’s local-contrast normalisation [31] and CLAHE [32] reduce illumination differences between cameras.

3 Materials

3.1 Datasets

IDRiD [6, 7] contains fundus photographs acquired in India with grades on the international five-level scale (0 no DR, 1 mild, 2 moderate, 3 severe non-proliferative, 4 proliferative DR). We use the publicly distributed Kaggle copy; after matching label files to images, [nidrid] graded images remain. **APTOS-2019** [16] provides [naptos] labelled images on the same scale from a different population and camera set; it is used only for pretraining. Table 1 gives the grade distributions.

3.2 Data partitioning

A stratified 15% of IDRiD ([ntest] images, random seed 42) is set aside as the test set before any training. It is not used for training, early stopping, model selection, threshold fitting or any hyper-parameter choice. The remaining [ntrainval] images are divided into five stratified folds. For APTOS pretraining a separate stratified 90/10 split is used for model selection.

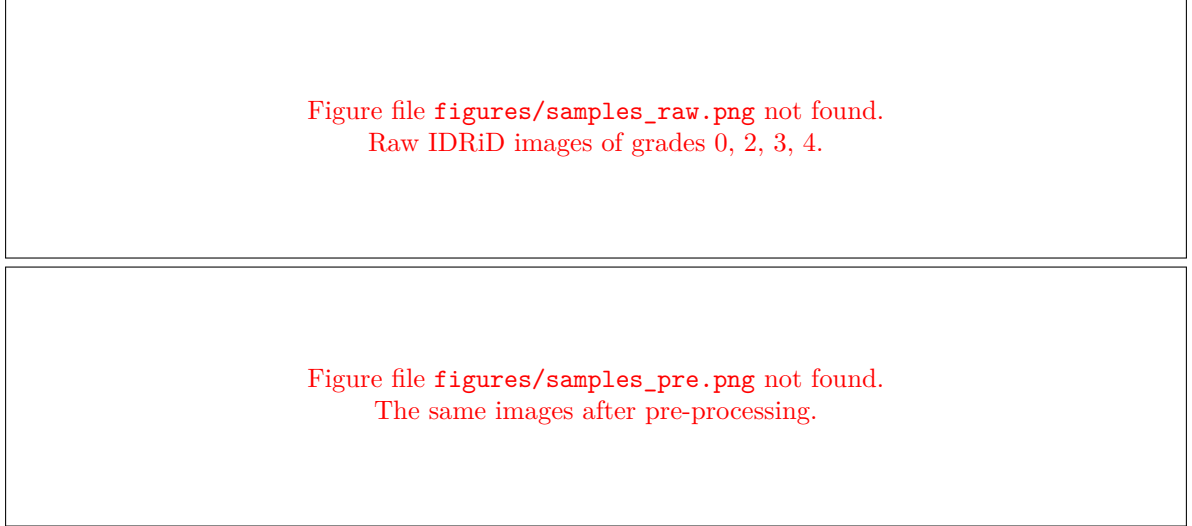


Figure 2: Example IDrID images before (top) and after (bottom) cropping, padding and Ben Graham normalisation.

3.3 Pre-processing

Each photograph is processed identically for both datasets (Fig. 2): images whose longer side exceeds 1600 pixels are first down-scaled; the black border is cropped (pixels with grey level > 10); the image is zero-padded to a square to preserve the aspect ratio of the fundus disc, and resized to 448×448 . We then apply Ben Graham’s local-contrast normalisation [31],

$$I_{\text{ben}} = 4I - 4(G_{\sigma} * I) + 128, \quad \sigma = 448/30, \quad (1)$$

where G_{σ} denotes Gaussian blurring, and set everything outside a centred circle of radius 0.48×448 pixels to grey (128). As an alternative we evaluate CLAHE [32] on the green channel (clip limit 2, 8×8 tiles) followed by bilateral filtering.

4 Method

Figure 3 gives an overview. Given an image $x \in \mathbb{R}^{3 \times 448 \times 448}$, a backbone produces a feature map that is turned into N tokens. Two experts process the tokens, exchange information, and are pooled and fused by a gate; two heads produce a class distribution and a continuous grade.

4.1 Backbone and tokenisation

We use EfficientNetV2-S [33] pretrained on ImageNet-21k and fine-tuned on ImageNet-1k [28], taken from the `timm` library [34]. Its last stage (stride 32, 256 channels) gives a 14×14 map for a 448×448 input. A 1×1 convolution with batch normalisation and GELU projects it to $F \in \mathbb{R}^{D \times h \times w}$ with $D = 256$, which is flattened into $N = hw = 196$ tokens $T \in \mathbb{R}^{N \times D}$.

4.2 Lesion-attention map and lesion expert

A light head (3×3 convolution to $D/4$ channels, GELU, 1×1 convolution to one channel, sigmoid) predicts a spatial map $m \in [0, 1]^{h \times w}$. No lesion annotation is used: m is learned end-to-end from grade labels and is encouraged to be sparse by an ℓ_1 penalty (Eq. 7). The lesion expert is the re-weighted token set

$$L = T \odot m, \quad L_i = m_i T_i, \quad i = 1, \dots, N. \quad (2)$$

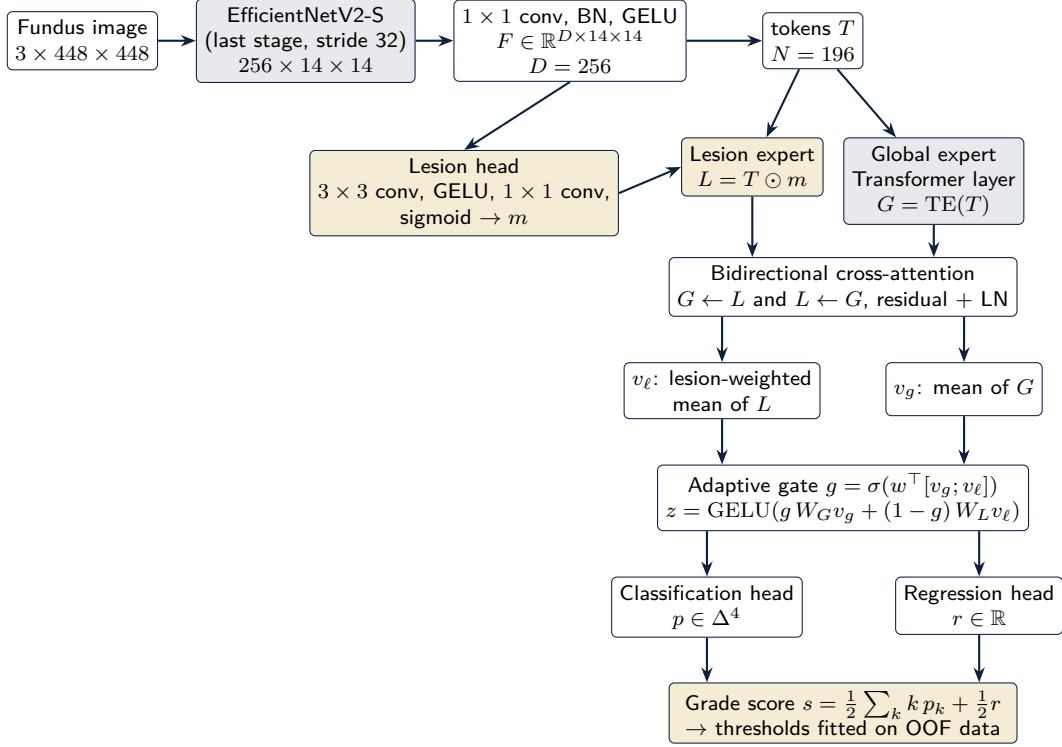


Figure 3: NITSCI architecture. Blue: backbone and global expert. Gold: lesion pathway and decision score. Token counts are for a 448×448 input. The lesion map m is learned from grade labels only.

4.3 Global expert

The global expert is one pre-norm Transformer encoder layer [22] with four heads, a feed-forward width of $2D$ and dropout 0.1, applied to all tokens: $G = \text{TE}(T)$. It models context across the whole retina, such as the spatial spread of lesions and their relation to the optic disc and macula.

4.4 Bidirectional cross-attention

Each expert queries the other with multi-head attention (four heads, dropout 0.1):

$$Z_g = \text{MHA}(Q=G, K=L, V=L), \quad G' = \text{LN}(G + Z_g), \quad (3)$$

$$Z_\ell = \text{MHA}(Q=L, K=G, V=G), \quad L' = \text{LN}(L + Z_\ell). \quad (4)$$

The global stream thus gathers lesion evidence, and the lesion stream receives the context needed to interpret it.

4.5 Pooling and adaptive gate

The global stream is mean-pooled, $v_g = \frac{1}{N} \sum_i G'_i$, and the lesion stream is pooled with the lesion weights, $v_\ell = \sum_i m_i L'_i / (\sum_i m_i + \epsilon)$. A scalar gate decides, per image, how much each expert contributes:

$$g = \sigma(w^\top [v_g; v_\ell] + b), \quad z = \text{GELU}(g W_G v_g + (1 - g) W_L v_\ell). \quad (5)$$

This is a two-expert soft mixture [14]; g is also reported for interpretability.

4.6 Dual head and grade score

Two heads share z : a classification head (LayerNorm, dropout 0.3, linear) gives logits and probabilities $p \in \Delta^4$, and a regression head of the same form gives $r = f_r(z) + 2$, centred on the middle

grade. Their outputs are combined into a continuous score

$$s = \frac{1}{2} \sum_{k=0}^4 k p_k + \frac{1}{2} r. \quad (6)$$

A grade is obtained by comparing s with four ordered thresholds $\theta_1 < \dots < \theta_4$: $\hat{y} = \#\{j : s \geq \theta_j\}$.

4.7 Training objective

For a mini-batch the loss is

$$\mathcal{L} = \text{CE}_{w,\varepsilon}(p, y) + \lambda_r \text{SmoothL1}(r, y) + \lambda_m \bar{m}, \quad (7)$$

where $\text{CE}_{w,\varepsilon}$ is cross-entropy with label smoothing $\varepsilon = 0.05$ [19] and class weights $w_c \propto n_c^{-1/2}$ (normalised to mean one, computed on the training fold), $\bar{m}$ is the mean of the up-sampled lesion map, $\lambda_r = 0.5$ and $\lambda_m = 0.01$. The square-root weighting is a compromise between ignoring and fully inverting the imbalance.

4.8 Two-stage training

Stage 1 (APTOS). The full model, initialised from ImageNet-21k backbone weights, is trained on APTOS-2019 for 10 epochs (learning rate 3×10^{-4} , EMA decay 0.999). **Stage 2 (IDRiD).** For each of the five folds, a model initialised from the stage-1 weights is fine-tuned for at most 25 epochs (learning rate 2×10^{-4} , EMA decay 0.99, early stopping with patience 8). In both stages we use AdamW [35] with weight decay 10^{-4} , a backbone learning rate one third of the head learning rate, a one-cycle cosine schedule with 10% warm-up [36], mixed precision [37], gradient clipping at norm 2, and an exponential moving average (EMA) of weights [30] that is used for validation and saved. The checkpoint with the highest sum of validation accuracy and quadratic weighted kappa (fixed thresholds 0.5, 1.5, 2.5, 3.5, no TTA) is kept.

Training augmentation: random horizontal and vertical flips, random rotation in $[-180^\circ, 180^\circ]$, translation up to 3%, scaling in $[0.95, 1.10]$, colour jitter (brightness and contrast 0.2, saturation 0.1, hue 0.02), and random erasing [38] with probability 0.2.

4.9 Inference and decision rule

Each fold model predicts every test image under four views (identity, horizontal flip, vertical flip, both); probabilities and regression outputs are averaged over views and over the five folds (20 predictions per image). The decision rule is chosen *on out-of-fold (OOF) predictions only* among three candidates: (i) argmax of p ; (ii) rounding s with thresholds 0.5, 1.5, 2.5, 3.5; (iii) optimised thresholds, found by coordinate search (four rounds, 80 candidate values per threshold, constrained to remain ordered) maximising OOF accuracy. The chosen rule was **[rule]**, with thresholds (**[thr]**).

Referable DR. For the screening task we use $P(\text{referable}) = p_2 + p_3 + p_4$ and select its operating threshold on OOF predictions by maximising Youden’s J [39] (selected value **[refthr]**).

4.10 Model size and cost

Table 2 lists the parameter counts and floating-point operations, measured with PyTorch [40] on a single 448×448 input. The cross-expert head adds 7% to the parameters of the backbone.

5 Experimental setup

5.1 Implementation

Experiments were run in PyTorch with `timm` and `scikit-learn` [41] on a single **[gpu]** GPU, batch size 16. Grad-CAM [42] visualisations use the `pytorch-grad-cam` library [43]. Random seeds are

Table 2: Model size and cost for one 448×448 image (FLOPs counted as $2 \times$ multiply-accumulates).

Component	Parameters (M)	Notes
EfficientNetV2-S feature extractor	19.85	256 output channels
Cross-expert head (all modules after the backbone)	1.40	$D = 256$, 4 heads
NITSCI total	21.25	23.0 GFLOPs
Plain baseline (same backbone, GAP, same dual head)	19.92	–
Inference latency per image (fp16, batch 16, [gpu]) [mspi] ms per model and view		

fixed (42, and $42 + k$ for fold k). The number of epochs actually run per fold after early stopping was (**[epochsrn]**).

5.2 Metrics

For five-class grading we report accuracy, balanced accuracy, quadratic weighted kappa (QWK) [26], Cohen’s kappa, Matthews correlation coefficient (MCC), macro-averaged precision, recall and F1, within-one-grade accuracy, macro one-vs-rest ROC AUC and average precision. For referable DR we report accuracy, sensitivity, specificity, positive and negative predictive value, F1 and ROC AUC.

5.3 Statistical analysis

95% confidence intervals are percentile bootstrap intervals over test images with 2000 resamples [44]. Paired comparisons between the full model and each ablation, which share the same test images, use the two-sided exact McNemar test on per-image correctness [45]. With **[ntest]** test images, differences of a few percentage points are within sampling error; we therefore interpret point estimates together with their intervals.

5.4 Compared configurations

All configurations below use identical splits, seeds, pre-processing (unless varied), augmentation and schedule:

- **Plain:** EfficientNetV2-S, global average pooling, the same dual head and loss (no experts).
- **w/o APTOS:** no stage-1 pretraining (ImageNet-21k initialisation only).
- **CLAHE:** CLAHE + bilateral filter instead of Ben Graham normalisation (APTOS stage repeated).
- **w/o ordinal head:** $\lambda_r = 0$.
- **w/o lesion map:** $m \equiv 1$, so the lesion expert sees all tokens equally.
- **w/o cross-attention:** the experts are not exchanged ($G' = \text{LN}(G)$, $L' = \text{LN}(L)$).
- **w/o gate:** fixed $g = 0.5$.

6 Results

6.1 Five-class grading

Table 3 reports the held-out test performance. Cross-validation estimates on the training portion are given for comparison: OOF accuracy **[oofacc]**% and QWK **[oofqwk]** (per-fold accuracy **[foldacc]**%, per-fold QWK **[foldqwk]**).

Table 4 and Fig. 4(a) show where errors occur. *[AUTHOR: Describe the confusion pattern: e.g. whether most errors are between adjacent grades (consistent with within-one-grade accuracy*

Table 3: Five-class DR grading on the held-out IDRiD test set ($n = \text{[ntest]}$). Ensemble of five fold models with four-view TTA, decision rule chosen on OOF data. Brackets: 95% bootstrap CI.

Metric	Value	Metric	Value
Accuracy (%)	[acc5] [acc5ci]	Macro precision (%)	[precmacro5]
Balanced accuracy (%)	[bacc5]	Macro recall (%)	[recmacro5]
Quadratic weighted kappa	[qwk5] [qwk5ci]	Macro F1 (%)	[f1macro5] [f1macro5ci]
Cohen’s kappa	[kappa5]	Weighted F1 (%)	[f1w5]
MCC	[mcc5]	Within-one-grade accuracy (%)	[within1]
Macro ROC AUC (OvR)	[auc5]	Macro average precision	[ap5]

Table 4: Per-grade precision, recall and F1 (%) on the test set.

	0 No DR	1 Mild	2 Moderate	3 Severe	4 PDR
Support	[nte0]	[nte1]	[nte2]	[nte3]	[nte4]
Precision	[p0]	[p1]	[p2]	[p3]	[p4]
Recall	[r0]	[r1]	[r2]	[r3]	[r4]
F1	[f0]	[f1]	[f2]	[f3]	[f4]

of **[within1]**%), and which grade is weakest. Grade 1 is typically hardest because it is defined by microaneurysms only.]

6.2 Referable DR screening

Table 5 reports the screening task. The operating threshold was fixed on OOF data before the test set was evaluated; the OOF AUC was **[oofrefauc]**. Figure 5 shows the ROC curves.

6.3 Ablation study

Table 6 compares the full model with the configurations of Section 5. All runs share the same test images, so the McNemar p -value tests whether per-image correctness differs from the full model.

[AUTHOR: Interpret Table 6 honestly: name the components whose removal lowers accuracy/QWK by more than the width of the bootstrap interval or with McNemar $p < 0.05$, and state plainly which components show no measurable effect at this test-set size. Do not claim a component “improves” performance if its p -value is large.]

6.4 Inference-time choices

Table 7 isolates the effect of the decision rule, TTA and fold ensembling without retraining. The decision rule was selected on OOF accuracy; test values are shown only to document the outcome of that choice.

6.5 Training behaviour

Figure 6 shows the training loss and validation metrics for each fold. *[AUTHOR: Comment on convergence, the epoch at which early stopping triggered, and the spread between folds.]*

6.6 Visual explanations

Figure 7 shows Grad-CAM maps for the predicted grade and the learned lesion-attention map m for test images, together with the gate value g (Eq. 5). Because m is learned without lesion annotations, these maps are qualitative only; we do not claim that they localise lesions. *[AUTHOR: Describe what the maps show, e.g. whether high attention falls on haemorrhages or exudates, and*

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Confusion matrices (5-class and referable).

Figure 4: Confusion matrices on the held-out test set. (a) Five-class grading. (b) Referable DR (grade ≥ 2).

Table 5: Referable DR (grade ≥ 2 vs. < 2) on the held-out test set. Brackets: 95% bootstrap CI.

Metric	Value		Metric	Value	
ROC AUC	[refauc]	[refaucci]	PPV (%)	[refppv]	
Sensitivity (%)	[refsens]	[refsensci]	NPV (%)	[refnpv]	
Specificity (%)	[refspec]	[refspecci]	F1 (%)	[reff1]	
Accuracy (%)	[refacc]	[refaccci]	TP / FN / FP / TN	[tp]	[fn] / [fp] / [tn]

whether g differs between low and high grades. A quantitative check against the IDRiD lesion masks (e.g. pointing-game accuracy) would strengthen this section.]

7 Discussion

7.1 Main findings

[AUTHOR: Summarise in three or four sentences, using only claims supported by Tables 3–7: (1) the absolute level of five-class and referable performance with CIs; (2) which ablations had the largest effect; (3) whether the ordinal head and threshold fitting reduced large errors (within-one-grade accuracy, QWK).]

The protocol choices matter as much as the architecture for a dataset of this size. Fitting thresholds on out-of-fold predictions rather than on the test set, and selecting the referable-DR operating point the same way, means that the reported test numbers are not optimistically biased by those choices. The cost is that the OOF estimates themselves are slightly optimistic, because each fold’s checkpoint was selected on the same fold whose predictions are later used for threshold fitting; the held-out test set is unaffected.

7.2 Comparison with published results

Direct numerical comparison with the IDRiD grading challenge [7] is not possible from this study, because the challenge uses an official train/test partition whereas we use a stratified random hold-out of the Kaggle copy. We therefore compare against a plain baseline and ablations trained under identical conditions (Table 6). [AUTHOR: Either (a) re-run the pipeline on the official IDRiD grading partition and add a table comparing with the challenge results reported in Porwal et al. [7] (copy the numbers from that paper and cite it), or (b) keep this paragraph as is.]

7.3 Limitations

- **Small test set.** With [ntest] test images, confidence intervals are wide and grade-level metrics for rare grades rest on few images.
- **Single target dataset.** All results are from IDRiD, which comes from one clinical centre. Performance on other populations and cameras (e.g. Messidor [20] or DDR [21]) is untested.

Figure file `figures/roc.pdf` not found.
ROC curves per grade and for referable DR.

Figure 5: One-vs-rest ROC curves for each grade and for referable DR (bold) on the test set.

Table 6: Ablation study on the held-out IDRiD test set. Five-class accuracy, QWK and macro F1; referable-DR AUC, sensitivity and specificity. *p*: exact McNemar test vs. the full model on five-class correctness.

Configuration	Params (M)	Acc (%)	QWK	F1 (%)	Ref. AUC	Sens. (%)	Spec. (%)
NITSCI (full)	[abfullparams]	[abfullacc]	[abfullqwk]	[abfullf1]	[abfullrefauc]	[abfullrefsens]	[abfullrefspec]
Plain (GAP head)	[abplainparams]	[abplainacc]	[abplainqwk]	[abplainf1]	[abplainrefauc]	[abplainrefsens]	[abplainrefspec]
w/o APTOS pretraining	[abnoaptosparams]	[abnoaptosacc]	[abnoaptosqwk]	[abnoaptosf1]	[abnoaptosrefauc]	[abnoaptosrefsens]	[abnoaptosrefspec]
CLAHE pre-processing	[abclaheparams]	[abclaheacc]	[abclaheqwk]	[abclahef1]	[abclaherefauc]	[abclaherefsens]	[abclaherefspec]
w/o ordinal head	[abnoregparams]	[abnoregacc]	[abnoregqwk]	[abnoregf1]	[abnoregrefauc]	[abnoregrefsens]	[abnoregrefspec]
w/o lesion map	[abnolesionparams]	[abnolesionacc]	[abnolesionqwk]	[abnolesionf1]	[abnolesionrefauc]	[abnolesionrefsens]	[abnolesionrefspec]
w/o cross-attention	[abnocrossparams]	[abnocrossacc]	[abnocrossqwk]	[abnocrossf1]	[abnocrossrefauc]	[abnocrossrefsens]	[abnocrossrefspec]
w/o adaptive gate	[abnogateparams]	[abnogateacc]	[abnogateqwk]	[abnogatef1]	[abnogaterefauc]	[abnogaterefsens]	[abnogaterefspec]

McNemar *p* vs. full: Plain **[abplainp]**; w/o APTOS **[abnoaptosp]**; CLAHE **[abclahep]**; w/o ordinal **[abnoregp]**; w/o lesion map **[abnolesionp]**; w/o cross-attention **[abnocrossp]**; w/o gate **[abnogatep]**. Parameter counts of “w/o cross-attention” and “w/o gate” include the unused modules, which are kept so that the remaining weights are initialised identically.

- **Non-official split.** The random hold-out prevents direct comparison with the IDRiD challenge.
- **Weakly supervised lesion map.** The lesion-attention map is not trained with lesion masks and has not been validated against them.
- **Single seed.** Variance from training randomness is reflected only through five folds, not through repeated runs with different seeds.
- **Reference standard.** Grades come from the dataset providers; inter-grader variability is not modelled.

7.4 Ethical and clinical considerations

The model is a research prototype and is not a medical device. Both datasets are public and de-identified. Before any clinical use, the system would need prospective, external validation across populations, cameras and image quality levels, with subgroup analysis and reporting according to TRIPOD+AI [46]. In screening, false negatives delay referral; the operating point should therefore be chosen together with clinicians to reach a target sensitivity rather than to maximise accuracy.

8 Conclusion

We presented NITSCI, a compact DR grading network that combines a global expert and a weakly supervised lesion expert through bidirectional cross-attention and an adaptive gate, with an ordinal dual head and a two-stage APTOS-to-IDRiD transfer protocol. Evaluated on a held-out IDRiD test set that was never used for any modelling decision, it reaches **[acc5]**% five-class accuracy (QWK **[qwk5]**) and a referable-DR AUC of **[refauc]**. *[AUTHOR: One sentence on the most important ablation finding.]* Future work will evaluate the model on the official IDRiD partition and on external cohorts, validate the lesion-attention maps against pixel-level annotations, and explore retinal foundation-model backbones [13].

Table 7: Effect of inference-time choices. Top: decision rules (OOF accuracy was used for selection). Bottom: single fold models vs. ensemble, with and without TTA (mean $\pm$ SD over five folds for single models).

Decision rule	OOF acc. (%)	Test acc. (%)	Test QWK
Argmax of p	[oofargmax]	[testargmax]	[testargmaxqwk]
Rounded score s	[oofrounded]	[testrounded]	[testroundedqwk]
Optimised thresholds on s	[oofopt]	[testopt]	[testoptqwk]

Prediction	Test acc. (%)	Test QWK
Single fold, TTA	[singleacc]	[singleqwk]
5-fold ensemble, no TTA	[notaacc]	[notaqwk]
5-fold ensemble + TTA	[ttaacc]	[ttaqwk]

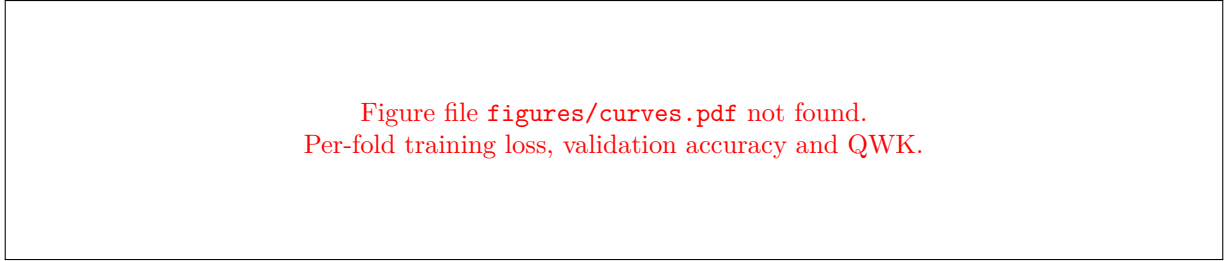


Figure 6: Training loss, validation accuracy and validation QWK per fold during IDRiD fine-tuning.

Declarations

CRedit authorship contribution statement. *[AUTHOR: e.g. Name: Conceptualization, Methodology, Software, Investigation, Writing – original draft. Name: Supervision, Writing – review & editing.]*

Declaration of competing interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability. IDRiD and APTOS-2019 are publicly available from their providers [6, 16]. The test-set indices, per-image predictions and all metrics are produced by the released code.

Code availability. *[AUTHOR: Repository URL (e.g. GitHub + Zenodo DOI) containing the notebook, export cell and trained weights.]*

Ethics statement. This study used only publicly available, de-identified datasets; no new data were collected from human participants.

Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work the authors used an AI assistant (Claude, Anthropic) to help draft and structure the manuscript text from the authors’ code. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. *[AUTHOR: Keep this statement accurate; most publishers, including Elsevier, require it.]*

Figure file `figures/explain.png` not found.
Original image, Grad-CAM, lesion attention for four test images.

Figure 7: Test images (top), Grad-CAM for the predicted grade (middle), and the learned lesion-attention map with gate value g (bottom), from the fold-1 model.

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